

Growth, Development, HDI, IHDI and MPI

From output expansion to capabilities, distribution and deprivation

01

Growth vs development

GROWTH: sustained rise in real GDP, real GVA or real income.

PER CAPITA: adjusts for population, not distribution.

DEVELOPMENT: structural transformation + capabilities + equity + agency.

SUSTAINABILITY: current gains must not destroy future welfare.

ANSWER LINE: growth expands means; development evaluates human ends.

MEANS -> CONVERSION -> HUMAN ENDS

02

Growth engine and quality

INPUTS: labour + physical capital + human capital + technology + institutions.

MECHANISM: capital deepening and productivity raise real output.

TRANSMISSION: jobs/wages + profits/investment + public revenue.

GOOD TRANSFORMATION: workers enter higher-productivity decent work.

FAILURES: jobless growth, enclaves, wage lag, regional gaps, pollution.

PRODUCTIVITY -> JOBS -> CAPABILITIES

03

HDI purpose and timeline

1990: first UNDP Human Development Report introduced HDI.

PURPOSE: compare average capability achievement, not economic size alone.

HEALTH: life expectancy at birth.

EDUCATION: expected years + mean years of schooling.

LIVING STANDARD: GNI per capita in PPP terms.

BOUNDARY: not poverty, inequality, freedom or sustainability by itself.

3 DIMENSIONS / 4 INDICATORS

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3 DIMENSIONS / 4 INDICATORS

04

HDI calculation decoder

2025 GOALPOSTS: LE 20-85 | EYS 0-18 | MYS 0-15.
INCOME: GNIpc 100-75,000 in constant 2021 PPP dollars.
LINEAR INDEX: $(\text{actual}-\text{min})/(\text{max}-\text{min})$.
EDUCATION: arithmetic mean of EYS and MYS indices.
INCOME: natural logs reflect diminishing capability gains.
FINAL: geometric mean limits compensation across dimensions.

$$\text{HDI} = (\text{HEALTH} \times \text{EDUCATION} \times \text{INCOME})^{(1/3)}$$

05

Interpretation firewall

VALUE: composite achievement on the stated method.
RANK: relative order among countries in that release.
VINTAGE: report year != reference year; component inputs may differ.
MOVEMENT: value can rise while rank falls if peers improve faster.
LIMITS: averages hide distribution; dimensions omit voice, safety and ecology.
GDI/GII: bounded gender complements, not extra HDI dimensions.

$$\text{RANK} \neq \text{VALUE} \neq \text{DATA YEAR}$$

06

IHDI inequality adjustment

METHOD: Atkinson-type inequality A_x for each HDI dimension.
ADJUSTMENT: $I\text{-adjusted} = (1-A_x) \times I$.
AGGREGATION: geometric mean of three adjusted dimensions.
EQUALITY: all $A_x=0 \rightarrow \text{IHDI}=\text{HDI}$; otherwise $\text{IHDI}<\text{HDI}$.
LOSS: $1-\text{IHDI}/\text{HDI}$; not a poverty rate or fourth dimension.
LIMIT: not association-sensitive across the same individuals.

$$I\text{-ADJUSTED}=(1-A)I; \text{IHDI} \leq \text{HDI}$$

07

Global MPI architecture

07

Global MPI architecture

INSTITUTIONS: OPHI + UNDP; Alkire-Foster method.

HEALTH 1/3: nutrition 1/6 + child mortality 1/6.

EDUCATION 1/3: years schooling 1/6 + attendance 1/6.

LIVING 1/3: fuel, sanitation, water, electricity, housing, assets.

Each living-standard indicator weighs 1/18.

TRAP: equal dimensions do not make all ten indicators equal.

3 DIMENSIONS / 10 INDICATORS

08

Dual cutoff and aggregation

CUTOFF 1: indicator-specific deprivation -> weighted score c.

CUTOFF 2: $c \geq 1/3$ identifies MPI poor.

CURRENT GLOBAL: vulnerable 1/5 to $<1/3$; severe $c \geq 1/2$.

H = poor population share; A = poor people's average score.

$MPI = H \times A$.

EXAMPLE: $H=.60$ and $A=.4167 \rightarrow MPI=.250$; H and A can diverge.

WHO IS POOR x HOW POOR

09

India's National MPI

NODAL PUBLISHER: NITI Aayog; DATA: NFHS household microdata.

12 INDICATORS: adds maternal health and bank account themes.

HEALTH: nutrition 1/6; mortality 1/12; maternal health 1/12.

EDUCATION: years schooling and attendance, each 1/6.

LIVING: seven indicators, each 1/21.

USE: national -> State/UT -> district and indicator diagnosis.

NATIONAL / STATE / DISTRICT

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Dated India evidence

REPORT: NITI Progress Review, released 17 July 2023.

DATA: NFHS-4 (2015-16) compared with NFHS-5 (2019-21).

HEADCOUNT: 24.85% -> 14.96%.

REPORT ESTIMATE: 12.5 crore people, critical multidimensional poverty

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REPORT: NITI Progress Review, released 17 July 2023.

DATA: NFHS-4 (2015-16) compared with NFHS-5 (2019-21).

HEADCOUNT: 24.85% -> 14.96%.

REPORT ESTIMATE: 13.5 crore people exited multidimensional poverty.

CAUTION: these are survey-period comparisons, not 2023 observations.

LIMIT: periodic survey and repeated cross-section create lag.

RELEASE YEAR != SURVEY PERIOD

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Comparison and limits

INCOME POVERTY: monetary command below a stated line.

MPI: direct, overlapping household deprivation.

HDI: average achievement | IHDI: inequality-adjusted achievement.

GLOBAL MPI 10 != INDIA NATIONAL MPI 12 indicators.

LIMITS: weights, cutoffs, threshold sensitivity, data lag.

HOUSEHOLD UNIT: useful for targeting, blind to some within-household gaps.

COMPLEMENTARY DASHBOARD

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Answer and policy spine

1 DEFINE the precise question and measure.

2 DECODE institution, unit, formula, threshold and dimensions.

3 DATE report year, data year and estimate status.

4 ANALYSE productivity -> jobs/services -> distribution -> environment.

5 QUALIFY averages, household unit, lag, sensitivity and causation.

6 PRESCRIBE productive inclusion, quality capabilities and resilience.

GROWTH -> FREEDOM, FAIRLY AND DURABLY

QUALIFIED VERDICT

No single rank is development. Use dated, disaggregated evidence to convert productive growth into broad capabilities without shifting costs to vulnerable people or future generations.